PADHAKU
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CHAPTER 14

Math of Space: Surface Area and Volume

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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🤔 Think and Reflect · 14.1 Cuboids and Cubes

1 **Why does the model of a cuboid as a stack of thin rectangular pieces (like a pack of cards) explain the formula volume = area of base × height?**

✔ ANSWER

Each thin piece is a rectangle of area l × w (the base area).

Stacking the pieces one on top of another up to height h fills the cuboid.

So the total space = (area of one piece) × (height of the stack) = l × w × h.

Volume = area of base × height = lwh.

2 **Write the formula for the surface area and the volume of a cube.**

✔ ANSWER

For a cube of side a (l = w = h = a):

Total surface area = $2(a^2 + a^2 + a^2)$ = $\mathbf{6a^2}$

Volume = a × a × a = $\mathbf{a^3}$

📝 Exercise Set 14.1

1 **The volume of a cube is 64 cm^3. What is its total surface area?**

✔ ANSWER

$a^3 = 64 \Rightarrow a = \sqrt[3]{64} = 4$ cm

TSA $= 6a^2 = 6 \times 4^2 = 6 \times 16$

TSA = 96 cm²

2 **How many small cubes with side 20 cm can be packed tight in a cubical box with side 2 m?**

✔ ANSWER

Side of box = 2 m = 200 cm

Cubes along each edge = 200 ÷ 20 = 10

Number of cubes = 10 × 10 × 10

1000 cubes

3 **The dimensions of a godown are 40 m × 25 m × 10 m. If it is filled with cuboidal boxes, each of dimensions 2 m × 1.25 m × 1 m, then find the number of boxes.**

✔ ANSWER

Along length: 40 ÷ 2 = 20

Along width: 25 ÷ 1.25 = 20

Along height: 10 ÷ 1 = 10

Number of boxes = 20 × 20 × 10 = 4000

(Check: 10000 m^3 ÷ 2.5 m^3 = 4000)

4000 boxes

4 **Two cubes each of volume 125 cm^3 are joined end to end. Find the surface area of**

the resulting cuboid.

✔ ANSWER

$a^3 = 125 \Rightarrow a = 5$ cm

Cuboid: l = 10 cm, w = 5 cm, h = 5 cm

TSA = 2(lw + wh + hl) = 2(50 + 25 + 50) = 2 × 125

250 cm²

5 **A cube of side 4 cm is cut into cubes of side 1 cm. What is the ratio of the surface areas of the original cube and all the cut-out cubes?**

✔ ANSWER

Original cube: SA = $6 \times 4^2 = 96$ cm²

Number of small cubes = $4^3 = 64$

SA of all small cubes = $64 \times 6 \times 1^2 = 384$ cm²

Ratio = 96 : 384

1 : 4

Volume stays 64 cm³, but the surface area becomes 4 times.

6 **The surface areas of the three faces of a cuboid that meet at one of the corners are 6 cm², 15 cm² and 10 cm². What is the volume of the cuboid?**

✔ ANSWER

Let the faces be lw = 6, wh = 15, hl = 10.

Multiply: $(lw)(wh)(hl) = (lwh)^2 = 6 \times 15 \times 10 = 900$

$lwh = \sqrt{900} = 30$

Volume = 30 cm³ (the cuboid is 2 cm × 3 cm × 5 cm)

7 **A cube of side 5 cm is painted on all its faces and sliced into 1 cm³ cubes. How many of these cubes have**
(i) exactly three faces painted? (ii) exactly two faces painted? (iii) exactly one face painted? (iv) no face painted?

✔ ANSWER

There are $5 \times 5 \times 5 = 125$ small cubes.

(i) Three faces: only the corner cubes ⇒ **8**

(ii) Two faces: on each of the 12 edges, leaving out 2 corners: $12 \times (5 - 2) =$ **36**

(iii) One face: on each of the 6 faces, the inner 3×3 cubes: $6 \times 9 =$ **54**

(iv) No face: the inner cube $3 \times 3 \times 3 =$ **27**

Check: $8 + 36 + 54 + 27 = 125$ ✓

8 **Find a cuboid with integer edges (in cm) with total surface area exactly 100 cm².**
(i) Is there more than one such cuboid? (ii) Can you find them all? (iii) Show that you have found them all.

✔ ANSWER

$2(lw + wh + hl) = 100 \Rightarrow lw + wh + hl = 50$.
Take $l \le w \le h$.

Example: $1 \times 2 \times 16$: $2 + 32 + 16 = 50$ ✓

(i) **Yes**, there is more than one.

(ii) **1 cm × 2 cm × 16 cm and 2 cm × 4 cm × 7 cm** ($2{\cdot}4 + 4{\cdot}7 + 7{\cdot}2 = 8 + 28 + 14 = 50$ ✓)

(iii) Proof: $lw + wh + hl = 50$ can be written as $(w + l)(h + l) = 50 + l^2$.

Since l is the smallest, $3l^2 \le lw + wh + hl = 50$, so $l \le 4$.

- $l = 1$: $(w + 1)(h + 1) = 51 = 3 \times 17 \Rightarrow w = 2, h = 16$
- $l = 2$: $(w + 2)(h + 2) = 54$, with $4 \le w + 2 \le h + 2 \Rightarrow 6 \times 9 \Rightarrow w = 4, h = 7$
- $l = 3$: $(w + 3)(h + 3) = 59$, a prime $\Rightarrow$ no cuboid
- $l = 4$: $(w + 4)(h + 4) = 66$, need $8 \le w + 4 \le \sqrt{66} \approx 8.1$, but 8 does not divide 66 $\Rightarrow$ none

So **only these two cuboids** exist.

🤔 Think and Reflect · Swimming pool

1 **A swimming pool is being built in your school. Choices: (A) 8 ft (depth) × 50 ft × 20 ft, (B) 6 ft (depth) × 100 ft × 15 ft.**
Which option would be suitable for your school? Why? What did you consider? [1 cubic foot ≈ 28.3 litres]

💡 SAMPLE ANSWER

Volume A = 8 × 50 × 20 = 8000 ft³ ≈ 8000 × 28.3 ≈ 2,26,400 litres

Volume B = 6 × 100 × 15 = 9000 ft³ ≈ 9000 × 28.3 ≈ 2,54,700 litres

Floor (land) area: A = 1000 ft², B = 1500 ft².

Option B suits a school better: it is shallower (6 ft, safer for students) and longer (100 ft, good for laps and races).

Option A needs less water and less land, but 8 ft is deep for beginners.

Aspects considered: safety (depth), space available, amount of water (cost, filling, cleaning), and the use (learning or racing).

⭐ 14.2 Right Circular Cylinder

? What is different in each of these cylinders (right circular and oblique circular)?

✔ ANSWER

In the right circular cylinder the axis is **perpendicular** to the circular base.

In the oblique circular cylinder the axis is **inclined (slanted)** to the base.

Both have circular cross-sections.

📝 Exercise Set 14.2

1 The radius of cylinder B is twice that of cylinder A, and the height of B is half that of

A. Find the ratio of the curved surface areas of A and B, and the ratio of their volumes.

✔ ANSWER

Let A have radius r, height h. Then B has radius 2r, height h/2.

CSA of A = $2\pi rh$; CSA of B = $2\pi(2r)(h/2) = 2\pi rh$

CSA ratio A : B = **1 : 1**

Volume of A = $\pi r^2 h$; Volume of B = $\pi(2r)^2(h/2) = 2\pi r^2 h$

Volume ratio A : B = **1 : 2**

2 **The radii of two cylinders are in the ratio 2 : 3 and their heights are in the ratio 3 : 2. Find (a) the ratio of their volumes, (b) the ratio of their curved surface areas.**

✔ ANSWER

Take radii 2r, 3r and heights 3h, 2h.

(a) $V_1 : V_2 = \pi(2r)^2(3h) : \pi(3r)^2(2h) = 12 : 18 =$ **2 : 3**

(b) $CSA_1 : CSA_2 = 2\pi(2r)(3h) : 2\pi(3r)(2h) = 6 : 6 =$ **1 : 1**

3 **The edge of a cube measures r cm. The largest possible right circular cylinder is cut out of the cube. What is the volume of the cylinder?**

✔ ANSWER

Largest cylinder: diameter = r, so radius = r/2; height = r.

$V = \pi(r/2)^2 \times r$

$V = \pi r^3/4$ cm³

4 **The radius of the base of a cylinder is increased by 10% and the height is decreased by x%. The volume remains unchanged. Find x.**

✔ ANSWER

New radius = 1.1r, new height = h(1 − x/100).

$\pi(1.1r)^2 \times h(1 - x/100) = \pi r^2 h$

$1.21 \times (1 - x/100) = 1 \Rightarrow 1 - x/100 = 1/1.21 = 100/121$

$x/100 = 21/121 \Rightarrow x = 2100/121$

x ≈ 17.36 (exactly 17 43/121)

5 **A solid metallic cube of side 12 cm is melted and recast into solid cylindrical rods, each of radius 2 cm and height 12 cm. Find:**

(i) the volume of the cube, (ii) the volume of one rod, (iii) the approximate number of complete rods. (π ≈ 22/7)

✔ ANSWER

(i) Volume of cube = 12^3 = **1728 cm³**

(ii) Volume of one rod = $\pi r^2 h = 22/7 \times 2 \times 2 \times 12 = 1056/7 \approx$ **150.86 cm³**

(iii) Number of rods = 1728 ÷ 150.86 ≈ 11.45

So **11 complete rods** can be made (some metal is left over).

📝 Exercise Set 14.3

1 **Find the total surface area of a cone, if its slant height is 21 m and diameter of its base is 24 m.**

✔ ANSWER

r = 12 m, l = 21 m

TSA = πr(l + r) = 22/7 × 12 × 33 = 8712/7

TSA ≈ 1244.57 m² (with π ≈ 3.14: ≈ 1243.44 m²)

2 **Find the curved surface area of a right circular cone whose slant height is 10 cm and base radius is 7 cm.**

✔ ANSWER

CSA = πrl = 22/7 × 7 × 10

CSA = 220 cm²

3 **The height of a cone is 16 cm and its base radius is 12 cm. Find the curved surface area and the total surface area.**

✔ ANSWER

l = √(h² + r²) = √(256 + 144) = √400 = 20 cm

CSA = πrl = 3.14 × 12 × 20 = **753.6 cm²** (= 240π)

TSA = $\pi r(l + r)$ = 3.14 × 12 × 32 = **1205.76 cm²** (= 384π)

4 **A cone has a height of 15 cm. If its volume is 1570 cm³, find the radius of the base.**

✔ ANSWER

$V = 1/3\ \pi r^2 h \Rightarrow 1570 = 1/3 \times 3.14 \times r^2 \times 15$

$1570 = 15.7\ r^2 \Rightarrow r^2 = 100$

r = 10 cm

5 **The curved surface area of a cone is 308 cm² and its slant height is 14 cm. Find (i) the radius of the base, (ii) the total surface area.**

✔ ANSWER

(i) $\pi rl = 308 \Rightarrow 22/7 \times r \times 14 = 308 \Rightarrow 44r = 308 \Rightarrow$ **r = 7 cm**

(ii) TSA = CSA + πr^2 = 308 + 22/7 × 49 = 308 + 154 = **462 cm²**

6 **A joker's cap is a right circular cone of base radius 7 cm and height 24 cm. Find the area of the sheet required to make 10 such caps.**

✔ ANSWER

$l = \sqrt{(24^2 + 7^2)} = \sqrt{(576 + 49)} = \sqrt{625} = 25$ cm

CSA of one cap = $\pi rl = 22/7 \times 7 \times 25 = 550$ cm²

For 10 caps = $10 \times 550 =$ **5500 cm²**

7 **What length of tarpaulin 3 m wide is required to make a conical tent of height 8 m and base radius 6 m? Extra 20 cm is needed for stitching and wastage.**

✔ ANSWER

$l = \sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$ m

CSA = $\pi rl = 3.14 \times 6 \times 10 = 188.4$ m²

Length = area ÷ width = $188.4 \div 3 = 62.8$ m

Add 20 cm = 0.2 m: $62.8 + 0.2$

63 m

8 **A right triangle with sides 6 cm, 8 cm and 10 cm is rotated through 360° about the side of 8 cm. Find the volume and the curved surface area of the solid formed.**

✔ ANSWER

The solid is a cone with h = 8 cm, r = 6 cm, l = 10 cm.

$V = 1/3\ \pi r^2 h = 1/3 \times \pi \times 36 \times 8 = 96\pi \approx 96 \times 3.14 =$ **301.44 cm³**

$CSA = \pi rl = \pi \times 6 \times 10 = 60\pi \approx$ **188.4 cm²**

9 **A cup is in the shape of a right circular cone. It is filled with water to half the depth of the**

cone. What fraction of the volume of the cup is occupied by the water?

✔ ANSWER

The cup stands with its tip at the bottom. The water forms a smaller cone of the same shape.

Its height is h/2, so (by similar triangles) its radius is r/2.

Water volume = $1/3\ \pi(r/2)^2(h/2) = 1/8 \times (1/3\ \pi r^2 h)$

Water fills 1/8 of the cup.

✏️ 14.5 Spheres · Try this

? **Find the connecting links between $V = 1/3 \times (4\pi r^2) \times r$ and the formula for the volume of a cone.**

✔ ANSWER

Divide the surface of the sphere into many tiny pieces and join each to the centre O.

Each piece becomes the base of a thin cone of height r.

Volume of each cone = 1/3 × (its base area) × r.

Adding all: volume of sphere = 1/3 × r × (sum of base areas) = $1/3 \times r \times 4\pi r^2$

$V = 4/3\ \pi r^3$

📝 Exercise Set 14.4

1 **A ball bearing has a radius of 0.7 cm. Find its surface area.**

✔ ANSWER

SA = $4\pi r^2$ = 4 × 22/7 × 0.7 × 0.7 = 4 × 22 × 0.07

SA = 6.16 cm^2

2 **Two solid spheres of the same metal weigh 5920 g and 740 g. Find the radius of the larger sphere, if the diameter of the smaller one is 5 cm.**

✔ ANSWER

Same metal ⇒ weight is proportional to volume.

$V_1 : V_2$ = 5920 : 740 = 8 : 1

$R^3 : r^3$ = 8 : 1 ⇒ R : r = 2 : 1

Smaller radius r = 5/2 = 2.5 cm ⇒ R = 2 × 2.5

R = 5 cm

3 **The diameter of the moon is about one fourth the diameter of the earth. Find the ratio of their surface areas.**

✔ ANSWER

Radius of moon = r, radius of earth = 4r.

SA ratio = $4\pi r^2 : 4\pi(4r)^2$ = 1 : 16

Moon : Earth = 1 : 16

4 **Find (i) the curved surface area and (ii) the total surface area of a hemisphere of radius 21 cm.**

✔ ANSWER

(i) CSA = $2\pi r^2$ = 2 × 22/7 × 21 × 21 = **2772 cm²**

(ii) TSA = $3\pi r^2$ = 3 × 22/7 × 21 × 21 = **4158 cm²**

5 **16 metal spheres, each of radius 2 cm, are packed in a box 16 cm × 8 cm × 8 cm. The box is then filled with preservative liquid. Find the volume of the liquid (to the nearest integer).**

✔ ANSWER

Volume of box = 16 × 8 × 8 = 1024 cm^3

Volume of one sphere = 4/3 π × 2^3 = 32π/3 cm^3

16 spheres = 512π/3 ≈ 512 × 3.1416 ÷ 3 ≈ 536.2 cm^3

Liquid = 1024 − 536.2 = 487.8

≈ 488 cm^3

6 **The radius of a sphere is increased by 10%. Show that the volume increases by about 33.1%.**

✔ ANSWER

New radius = 1.1r

New volume = $4/3\ \pi(1.1r)^3 = 1.331 \times 4/3\ \pi r^3$

Increase = 1.331 − 1 = 0.331 of the old volume

Increase = 33.1% (shown)

7 **The radius of a sphere is increased by x%. The volume increases by 72.8%. Find x.**

✔ ANSWER

$(1 + x/100)^3 = 1.728$

$1 + x/100 = \sqrt[3]{1.728} = 1.2$ (since $12^3 = 1728$)

x = 20

8 **The circumference of the base of a hemispherical dome is 35.2 m. Find the cost of painting it at ₹10 per 100 cm².**

✔ ANSWER

$2\pi r = 35.2 \Rightarrow r = 35.2 \times 7 \div 44 = 5.6$ m

CSA = $2\pi r^2 = 2 \times 22/7 \times 5.6 \times 5.6 = 197.12$ m²

1 m² = 10000 cm² ⇒ 197.12 m² = 19,71,200 cm²

Cost = 19,71,200 ÷ 100 × 10 = **₹1,97,120**

⭐ 14.6 Areas and Volumes Around Us

1 **Identify and list objects around you that are cubical, cuboidal, cylindrical, conical, spherical or pyramidal (or nearly so).**

💡 SAMPLE ANSWER

Shape	Examples
Cube	dice, Rubik's cube, sugar cube
Cuboid	brick, book, almirah, matchbox, room
Cylinder	pipe, gas cylinder, glass, drum, pencil
Cone	ice-cream cone, birthday cap, funnel, tent
Sphere	ball, marble, orange, globe
Pyramid	tent roofs, temple tops, Egyptian pyramids

2 **Activity: Find the approximate volumes of a gas cylinder, a bucket, a cylindrical vessel, a cupboard and a water tank near you (1000 cm^3 = 1 litre).**

💡 **SAMPLE ANSWER**

Measure, treat each as a simple solid, then use the formula. Sample values:

- Gas cylinder (r ≈ 15 cm, h ≈ 60 cm): $\pi r^2 h \approx 3.14 \times 225 \times 60 \approx 42{,}390\ cm^3 \approx 42$ L
- Bucket (r ≈ 14 cm, h ≈ 30 cm, as a cylinder): $\approx 18{,}500\ cm^3 \approx 18$ L
- Cupboard (180 × 90 × 50 cm): $8{,}10{,}000\ cm^3 \approx 810$ L
- Water tank (r ≈ 55 cm, h ≈ 110 cm): $\approx 10{,}45{,}000\ cm^3 \approx 1000$ L

Your answers depend on your own measurements.

? **Example 6: Make an estimate of the number of tennis balls (radius 3 cm) in the classroom**

using the second and third configurations.

💡 SAMPLE ANSWER

Upper bound (no gaps): ≈ 33,45,132 balls. The first (simple) packing fills about 52% of space and gave ≈ 16,87,500.

In configurations B and C each layer sits in the hollows of the layer below, so layers are closer and gaps are smaller.

- B fills about 68% of space: 33,45,132 × 0.68 ≈ **22.7 lakh balls**
- C (closest packing) fills about 74%: 33,45,132 × 0.74 ≈ **24.8 lakh balls**

So roughly 22–25 lakh balls, more than the first arrangement.

📝 End-of-Chapter Exercises

1 **Estimate how many scoops of ice cream can be obtained from a cuboidal container 10 cm × 15 cm × 20 cm.**

💡 SAMPLE ANSWER

Volume of container = 10 × 15 × 20 = 3000 cm^3

Assume a scoop is a ball of radius 2.5 cm: 4/3 × 3.14 × 2.5^3 ≈ 65 cm^3

Scoops ≈ 3000 ÷ 65 ≈ 46

About 45 scoops

2 **A cube of integer side a is made of unit cubes. Write an expression for the number of unit cubes to be added to make a cube of side a + 1.**

✔ ANSWER

Needed = $(a + 1)^3 - a^3 = a^3 + 3a^2 + 3a + 1 - a^3$

$3a^2 + 3a + 1$

3 (i) **Could a person drink enough water in a lifetime to fill an entire room the size of your classroom?**

💡 SAMPLE ANSWER

Assume classroom 8 m × 6 m × 3.5 m = 168 m^3 = 1,68,000 L.

A person drinks about 2.5 L a day for 70 years: 2.5 × 365 × 70 ≈ 64,000 L.

No, only about 40% of the room would be filled.

3 (ii) **Estimate the number of bricks used to build the walls of your classroom.**

💡 SAMPLE ANSWER

Assume room 8 m × 6 m, walls 3.5 m high and 0.23 m thick.

Wall area = 2(8 + 6) × 3.5 = 98 m^2; less door and windows ≈ 6.5 m^2 ⇒ 91.5 m^2

Wall volume ≈ 91.5 × 0.23 ≈ 21 m^3

One brick with mortar ≈ 25 cm × 12.5 cm × 7.5 cm ≈ 0.00234 m^3

Bricks ≈ 21 ÷ 0.00234 ≈ **9000 bricks**

4 **Options: (A) one chocolate cube of side 50 mm, (B) hundred chocolate cubes of side 10 mm. Which gives more chocolate?**

✔ ANSWER

(A) 50^3 = 1,25,000 mm^3

(B) 100 × 10^3 = 1,00,000 mm^3

Option A gives more chocolate.

5 **If all the people in the world are crowded together in one location, how much area would it cover?**

💡 SAMPLE ANSWER

Assume about 8 billion people, each standing in about 0.5 m × 0.5 m = 0.25 m^2.

Area ≈ 8,00,00,00,000 × 0.25 = 2,00,00,00,000 m^2 = 2000 km^2

About 2000 km^2, a square of side about 45 km.

6 **Milk glasses are cylinders of diameter 7 cm, filled to 12 cm. How many litres of milk are needed for 1600 students?**

✔ ANSWER

r = 3.5 cm, h = 12 cm

One glass = $\pi r^2 h$ = 22/7 × 3.5 × 3.5 × 12 = 462 cm^3

1600 glasses = 462 × 1600 = 7,39,200 cm^3

÷ 1000 ⇒ **739.2 litres**

7 **The surface area of a sphere of radius 5 cm is five times the curved surface area of a cone of radius 4 cm. Find the height and the volume of the cone.**

✔ ANSWER

SA of sphere = $4\pi \times 5^2 = 100\pi$

CSA of cone = $100\pi \div 5 = 20\pi \Rightarrow \pi \times 4 \times l = 20\pi \Rightarrow l = 5$ cm

$h = \sqrt{(l^2 - r^2)} = \sqrt{(25 - 16)}$ = **3 cm**

$V = 1/3\ \pi \times 16 \times 3 = 16\pi \approx$ **50.29 cm^3**

8 **Earth radius 6370 km, Jupiter 69,900 km, Sun 6,95,700 km. Find approximately (i) volume of Jupiter : volume of Earth, (ii) volume of Sun : volume of Earth.**

✔ ANSWER

Volume ratio = (radius ratio)³ (4/3 π cancels).

(i) 69900 ÷ 6370 ≈ 10.97; $10.97^3 \approx 1321$ ⇒ **about 1320 : 1**

(ii) 695700 ÷ 6370 ≈ 109.2; $109.2^3 \approx 13{,}03{,}000$ ⇒ **about 13 lakh : 1** (1.3 million)

9 **Show that the volume of a sphere is 2/3 of the volume of the smallest cylinder which encloses it.**

✔ ANSWER

Smallest enclosing cylinder: radius r, height 2r.

Cylinder volume = $\pi r^2 \times 2r = 2\pi r^3$

Sphere volume = $4/3\ \pi r^3 = 2/3 \times 2\pi r^3$

Sphere = 2/3 × cylinder (shown)

10 **A string is wrapped tightly around the Earth's equator. Another string 1 m longer forms a larger circle, the same height above the ground everywhere. How high is it?**

Repeat for (i) the Moon (ii) Jupiter (iii) a volleyball. Are you surprised?

✔ ANSWER

Let the radius be R and the gap be d.

$2\pi(R + d) - 2\pi R = 1 \Rightarrow 2\pi d = 1 \Rightarrow d = 1/(2\pi)$ m

$d \approx 0.159$ m $\approx$ **16 cm**

The answer does not depend on R, so for (i) the Moon, (ii) Jupiter and (iii) a volleyball the gap is also **about 16 cm**.

It is surprising at first, because 1 m seems tiny compared with the Earth, but the extra length needed depends only on the gap ($2\pi d$), not on the size of the sphere.

11 **A square pyramid is fitted inside a cylinder (radius r, height h): its base corners on the base circle, its apex on the top. Find the ratio of the volume of the pyramid to the volume of the cylinder.**

✔ ANSWER

The diagonal of the square base = diameter = 2r.

Area of square = (diagonal)² ÷ 2 = (2r)² ÷ 2 = $2r^2$

Pyramid volume = 1/3 × $2r^2$ × h = $2r^2h/3$

Cylinder volume = $\pi r^2 h$

Ratio = ($2r^2h/3$) : $\pi r^2 h$ = **2 : 3π** (≈ 0.21)

12 (i) **Change in volume when the length of a cuboid l, w, h is increased by 1 unit?**

(a) 1 (b) lwh + 1 (c) wh (d) lw (e) lh cubic units

✔ ANSWER

(l + 1)wh − lwh = wh

(c) wh cubic units

12 (ii) **Change in volume when the radius of a cylinder r, h is increased by 1 unit?**

(a) 1 (b) $\pi r^2 h$ (c) πr^2 (d) 2πrh + 2πh (e) 2πrh + πh cubic units

✔ ANSWER

$\pi(r+1)^2h - \pi r^2h = \pi(2r+1)h = 2\pi rh + \pi h$

(e) $2\pi rh + \pi h$ cubic units

12 (iii) **Change in volume when the radius of a sphere is decreased by 1 unit?**

✔ ANSWER

$4/3\ \pi r^3 - 4/3\ \pi(r-1)^3 = 4/3\ \pi(3r^2 - 3r + 1)$

Decrease = $4\pi r^2 - 4\pi r + 4\pi/3$ cubic units

13 **Given a cube with volume V, express its total surface area S in terms of V.**

✔ ANSWER

$a = \sqrt[3]{V} = V^{(1/3)}$

$S = 6a^2$

$S = 6V^{(2/3)}$ $(= 6\sqrt[3]{V^2})$

14 **Given a cube with total surface area S, express its volume V in terms of S.**

✔ ANSWER

$6a^2 = S \Rightarrow a = \sqrt{(S/6)}$

$V = a^3$

$V = (S/6)^{(3/2)}$ $(= (S/6)\sqrt{(S/6)})$

15 **A cylindrical glass (height 25 cm, radius 4 cm) has water up to 16 cm. The water must**

reach 20 cm for the crow. How many marbles of radius 1 cm should the crow drop in?

✔ ANSWER

Rise needed = 20 − 16 = 4 cm

Volume to add = $\pi \times 4^2 \times 4 = 64\pi$ cm³

One marble = $4/3\ \pi \times 1^3 = 4\pi/3$ cm³

Marbles = $64\pi \div (4\pi/3) = 48$

48 marbles

16 **Find the volume of ink in a new ball point pen. Take measurements and make approximations.**

💡 SAMPLE ANSWER

Treat the ink in the refill as a thin cylinder.

Assume inner radius ≈ 1 mm = 0.1 cm and ink length ≈ 10 cm.

$V = \pi r^2 h \approx 3.14 \times 0.01 \times 10 \approx 0.31$ cm³

About 0.3 cm³ (0.3 mL) of ink

17 (i) **A ball of chapati dough has radius 6 cm. Estimate how many chapatis can be made from it.**

💡 SAMPLE ANSWER

Dough volume = $4/3\ \pi \times 6^3 = 288\pi$ cm³

Assume a chapati is a thin cylinder: radius 8 cm, thickness 0.2 cm ⇒ $\pi \times 64 \times 0.2 = 12.8\pi$ cm³

Number = 288π ÷ 12.8π = 22.5

About 22 chapatis

17 (ii) **Cut a coconut/muskmelon in half and find the approximate volume of edible flesh by taking measurements.**

💡 **SAMPLE ANSWER**

Treat half the coconut flesh as a hollow hemisphere.

Sample: outer radius of flesh R = 6 cm, inner radius r = 5 cm.

Volume = $2/3\ \pi(R^3 - r^3) = 2/3 \times 3.14 \times (216 - 125) \approx 190\ cm^3$

About 190 cm^3 in each half (about 380 cm^3 in the whole coconut)

18 (i) **If all the pages of this textbook were laid out side by side on the floor, would they cover the entire classroom floor?**

💡 **SAMPLE ANSWER**

A page is about 20 cm × 27 cm ≈ 0.054 m^2.

Assume about 170 pages, all laid flat (one side each): $170 \times 0.054 \approx 9.2\ m^2$

Classroom floor ≈ 8 m × 6 m = 48 m^2

No, they would cover only about one fifth of the floor.

18 (ii) **What is the maximum number of textbooks that can fit in an empty storeroom of 15 ft × 20 ft × 30 ft?**

💡 **SAMPLE ANSWER**

Room = 9000 ft^3 ≈ 9000 × 28,317 ≈ 25,48,53,000 cm^3

Book ≈ 27 cm × 20 cm × 1 cm = 540 cm^3

Books ≈ 25,48,53,000 ÷ 540 ≈ 4,72,000

About 4.7 lakh textbooks (if packed with no gaps)

19 **If the entire human population climbed into one giant cube, how long would the side have to be?**

💡 **SAMPLE ANSWER**

Assume 8 billion people, each about 0.07 m^3 (70 kg, about the density of water).

Total volume ≈ 8 × 10^9 × 0.07 = 5.6 × 10^8 m^3

Side = $\sqrt[3]{(5.6 \times 10^8)}$ ≈ 824 m

About 0.8 km (more if we allow gaps between people)

20 **1.38 billion km³ of water forms a uniform layer over the Earth (radius ≈ 6371 km). Estimate the thickness.**

(i) Write an expression for the thickness. (ii) Simplify it using a calculator.

✔ ANSWER

(i) The layer is thin, so volume ≈ surface area × thickness:

$t \approx V \div (4\pi R^2) = 1.38 \times 10^9 \div (4\pi \times 6371^2)$ km

(Exact: $4/3\ \pi[(R + t)^3 - R^3] = V \Rightarrow t = \sqrt[3]{(R^3 + 3V/(4\pi))} - R$)

(ii) $4\pi \times 6371^2 \approx 51{,}00{,}64{,}000$ km²

$t \approx 1{,}38{,}00{,}00{,}000 \div 51{,}00{,}64{,}000 \approx 2.7$ km

About 2.7 km thick

21 **Give the dimensions of a cuboid whose volume is halved when its surface area is doubled.**

💡 SAMPLE ANSWER

Reading it as: the number for the volume is half the number for the surface area ($V = S \div 2$).

Cube 3 × 3 × 3: $V = 27$, $S = 6 \times 9 = 54 \Rightarrow 27 = 54 \div 2$ ✓

Other answers: 4 × 4 × 2 ($V = 32$, $S = 64$), 2 × 3 × 6 ($V = 36$, $S = 72$).

For example, 3 cm × 3 cm × 3 cm

22 **Project: Find the volume of your house making necessary approximations. Present how you solved it.**

💡 **SAMPLE ANSWER**

Steps:

- Treat each room as a cuboid; measure length, width and height (pace or tape).
- Volume of each room = l × w × h; add all rooms (kitchen, toilet, verandah too).

Sample: 2 rooms 4 m × 3 m × 3 m = 36 m^3 each, kitchen 3 m × 2 m × 3 m = 18 m^3, toilet 2 m × 1.5 m × 3 m = 9 m^3

Total ≈ 36 + 36 + 18 + 9 = **99 m^3 ≈ 100 m^3**

Mention the approximations (walls ignored, ceiling height taken as equal).

🎉 **Chapter 14 done! Revise the answers once more tomorrow.**